School of Physics,
Engineering and
Computer Science

# MSc Data Science Project
# 7PAM2002

Department of Physics, Astronomy and Mathematics

## Data Science FINAL PROJECT REPORT

**Project Title:**

The Art of Convincing: Can AI Out-Persuade Humans?

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# DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

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## ABSTRACT

This report explores whether artificial intelligence can be equally persuasive to humans through the attributes of claims and arguments that can drive both initial and final persuasion ratings, and the degree to which prompt types and sources can drive these ratings. The study aimed to quantify the persuasion shift, the textual and contextual influence of successful persuasion, and contrast AI-generated and human-written arguments in a systematic analysis. The Anthropic Persuasion dataset was used to implement a quantitative research design; the dataset includes claims, arguments, source labels, prompt types, and pre- and post-exposure stance ratings. The conversion to food ratings, text cleanup, and feature engineering were performed, and predictive performance was evaluated using exploratory analysis, Regression modelling, and machine learning. Results indicate that persuasion shift was overall small. The initial standpoint was the most prominent indicator of the ultimate rating, with the source's characteristics, the type of prompt, and arguments also playing lesser yet significant roles. The report suggests that AI can be as close as possible to human persuasion performance, but it will be effective only conditionally, depending on the situation and the audience's predispositions.

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# CHAPTER 1: INTRODUCTION

### *1.1 Background*

Persuasion is highlighted in relation to health, purchases, or policy-making; in general, it is not a slight matter and is aimed at transforming people's opinions rather than merely pouring information on them. With big language models now providing advice, explanations, and arguments, the question has become an empirical question of its own: whether AI arguments can persuade people just as successfully as human arguments, and in what circumstances (Fischetti, 2025). Anthropic's Persuasion Dataset was deployed to test precisely this: short policy-relevant claims were paired with arguments generated by a human or a model, and the variation in agreement with those arguments after reading them was observed. The experiment has the participants rate their position on a claim first, read an argument, and rate afterwards. The data consists of the starting and finishing ratings, and metadata on the author of the argument (e.g., a human or a particular model), as well as the type of prompt usedto produce the amodel's arguments Their breakdown reveals that persuasiveness is projected to improve with more recent model generations, and that Claude 3 Opus generated arguments that were statistically identical to human ones in their experiment (Barale, 2021). This preconditions the analysis of which features of claims and arguments cause persuasion and whether some prompting strategies equally enhance persuasion.

### *1.2 Aim and Objectives*

The study aims to determine whether an AI-based argument can be as persuasive and effective as an argument a human would write, and to identify the elements of an argument and prompts that drive those changes in attitude. The objectives of the study are:

- To measure persuasion changes by calculating the shift from rating_initial to rating_final across argument sources and prompt_type conditions.
- To identify what drives persuasion by testing which claim and argument features (textual and contextual) predict higher final ratings and larger rating shifts.
- To compare prompting strategies to determine which prompt types produce the most persuasive AI arguments relative to human arguments.

### *1.3 Research Question*

- What are the characteristics of claims and arguments that influence their initial and final persuasiveness ratings, and how do different prompt types and sources (e.g., AI models like Claude) affect these metrics?

### *1.4 Rationale of the Study*

The study investigates persuasion as a machine-learning problem using the Anthropic Persuasion Dataset, which includes various metrics such as claims, arguments, and persuasiveness scores. It employs Linear Regression to analyze predictor strength and Random Forest to determine influential variables based on structured and textual features. Additionally, AI models (unsloth/tinyllama-chat-bnb-4bit and unsloth/Llama-3.2-1B-bnb-4bit) will be fine-tuned to generate or score persuasive arguments. The effectiveness of AI-generated persuasion will be evaluated against human and Claude-based benchmarks, aiming to assess the explainability and measurability of AI persuasion (Dehnert and Mongeau, 2022; Soliman, 2024).

# CHAPTER 2: LITERATURE REVIEW

## *2.1 Introduction*

Chapter 2 situates current research within an academic framework that views persuasion as the use of communicative information to influence beliefs, emotions, or actions. It highlights how social psychology and communication studies have employed dual-process theories to differentiate between deliberative and heuristic processing. The rise of big language models and conversational AI extends the discussion to how these technologies can generate and convey messages that influence human cognition. The chapter emphasizes rigorous empirical research, focusing on peer-reviewed studies that evaluate persuasive efficacy through controlled trials contrasting AI-mediated and human-mediated persuasion, aiming to establish a solid strategy for investigating AI in persuasive contexts.

## *2.2 Review of Relevant Studies*

### *2.2.1 Paper 1: Lin et al. (2025) — Persuading voters using human–AI dialogues (Nature)*

Lin *et al.* (2025) provide one of the first empirical measures of what AI systems can do to modify political opinions through debate. The study investigated the impact of interactions with a large-scale language model on voter preferences for the 2024 US presidential and 2025 Canadian and Polish elections. Participants were randomly assigned to assess changes in preferences due to AI dialogues. Significant shifts in voter sentiment were found, often surpassing traditional media effects. The research indicates that AI models prioritize factual accuracy and persuasion over psychological manipulation. While right-wing views showed more deception, they varied in authenticity. The study underscores the influence of AI on attitudes in realistic decision-making contexts but limits its implications to political scenarios, lacking broader applications to areas like health or policy. Future research is needed to explore motivational arguments and their variability based on prompts and models.

### *2.2.2 Paper 2: Dehnert and Mongeau (2022) — Persuasion in the age of artificial intelligence (Human Communication Research)*

Dehnert and Mongeau (2022) reclassify persuasion in the age of AI rather than quantifying its effects. This definition addresses artificial systems as active communicators, proposing an evolution of traditional persuasion models through CASA, MAIN, and heuristic-systematic perspectives. It highlights how AI embodiment influences human message perception, emphasizing that AI-generated arguments may elicit different reactions compared to human-generated ones. Dehnert and Mongeau's (2022) theoretical framework support this notion by illustrating the impact of source cues and context on message processing. The article distinguishes itself by explaining the role of AI agents in persuasion rather than focusing on quantitative outcomes. Its comprehensive analysis melds classical communication theories with human-AI interaction, paving the way for nuanced empirical interpretations, while noting the absence of quantitative data or text-level analysis. The study's assumptions and analysis are contextualized, suggesting that its value could be enriched by further persuasive and literary research.

### *2.2.3 Paper 3: Burtell and Woodside (2023) — Artificial Influence: An Analysis of AI-Driven Persuasion*

Burtell and Woodside (2023) study AI and persuasion. Without empirical evidence, the writers explore AI's potential to influence societal persuasion. Advanced AI systems affect user behavior through personalized content and dialogue, suggesting that AI's capacity for humor and customization may reshape digital beliefs and decision-making. The framework presented highlights how chatbots, recommendation engines, and strategic game agents can alter discourse and information ecology. However, Burtell and Woodside caution about the risks of AI-generated messages misleading users and threatening autonomy. The research emphasizes the importance of linguistic features, personalization, and context in persuasive messaging. It raises questions about the integrity of AI-influenced communication and its long-term consequences, arguing for further exploration of AI's persuasive capabilities amidst the challenges of empirical modeling without controlled results.

#### *2.2.4 Paper 4: Millet et al. (2023) — Defending humankind: Anthropocentric bias in the appreciation of AI art*

Millet *et al.* (2023) found that an anthropocentric worldview influences how creative outputs from AIs and humans are assessed. The study included four studies with 1,708 participants who saw repeated paintings and music originating from AI and vice versa. Those who thought the makers were AI assessed the works poorly and felt less amazement than those who thought they were humans, especially those with high human creativity ratings. Motivated reasoning hinders objective aesthetic evaluations and emphasises a human-centered worldview, the authors noted. This prejudice may affect how AI-generated persuasive texts are evaluated, lowering AI products' persuasiveness. Due to its excellent experimental design and pre-registered sample, the study shows the impacts of source label manipulation. The results resemble persuasion studies but focus on art judgement rather than persuasive mechanisms or text characteristics' influence on attitude change. Research mentions AI output biases. However, the paper does not specify which text or rhetorical elements may influence data-driven persuasion.

### ***2.3 Key Challenges and Research Gaps***

This study is influenced by several obstacles and shortcomings. A large-scale political conversation in which conversational bots changed voters shows that AI systems can affect human beliefs. The literature suggests that persuasive messages are interpreted differently depending on variables, however this requires an independent investigation of persuasive mechanisms. Psychological variables including audience biases against AI outputs reduce persuasion efficacy, especially when participants know AI produced them. Despite evidence that AI can influence views, little research has linked textual elements to persuasion effects. This study will evaluate how evidence density, framing, source signals, and prompts affect text. It affects persuasiveness, and empirical and theoretical answers based on message quality and audience interpretation agree.

# CHAPTER 3: DATASET

## *3.1 Dataset Description*

The Persuasion Dataset from Hugging Face's "Anthropic/persuasion" homepage, detailed in the April 9, 2024 research note "Measuring the Persuasiveness of Language Models," was developed by Esin Durmus and colleagues to evaluate the persuasiveness of large language models. It comprises 56 opinionated assertions formed by combining 28 policy concerns with supporting and opposing claims. In a study with 3,832 U.S. participants, three individuals crafted ~250-word persuasive messages for each claim using four argument styles: Compelling Case, Role-playing Expert, Logical Reasoning, and Deceptive. Pre/post ratings measured shifts in agreement, allowing for comparison of AI versus human persuasive effectiveness. This dataset is notable for its structured link between argument language, source, and prompt type, aligning with the research goals.

| Feature | What it represents | Typical use in analysis |
|---|---|---|
| worker_id | Anonymous identifier for the participant who provided both the pre- and post-stance ratings | Group/repeated-measures checks (e.g., controlling for participant-level tendencies) |
| claim | The policy-relevant statement that the participant is asked to evaluate | Topic/issue controls; claim-level difficulty or contentiousness |
| argument | The persuasive text presented to the participant (written by a human or generated by a language model) | Text-feature extraction (tone, evidence, length, framing) |
| source | The author/source of the argument (a model name or "Human") | Comparing human vs AI (and model-to-model) persuasive performance |
| prompt_type | The prompting strategy used to generate the model argument | Testing which prompting strategies yield larger attitude shifts |
| rating_initial | Participant's stance/agree–disagree rating before reading the argument | Baseline attitude: covariate for predicting susceptibility to persuasion |
| rating_final | Participant's stance/agree–disagree rating after reading the argument | Outcome for persuasion; used to compute change (rating_final − rating_initial) |

## *3.2 Exploratory Data Analysis*

The exploratory analysis begins by measuring whether attitudes changed after participants read an argument, then compares those changes across prompt types and argument sources. This structure links the figures directly to the main research question: whether persuasion is mainly explained by the argument itself, the source that produced it, or the participant's starting position. Persuasion shift was calculated as rating_final minus rating_initial. This metric is useful because it converts the before-and-after ratings into a direct measure of attitude movement: positive values indicate movement towards stronger agreement with the claim, negative values indicate backfire, and zero indicates no change.

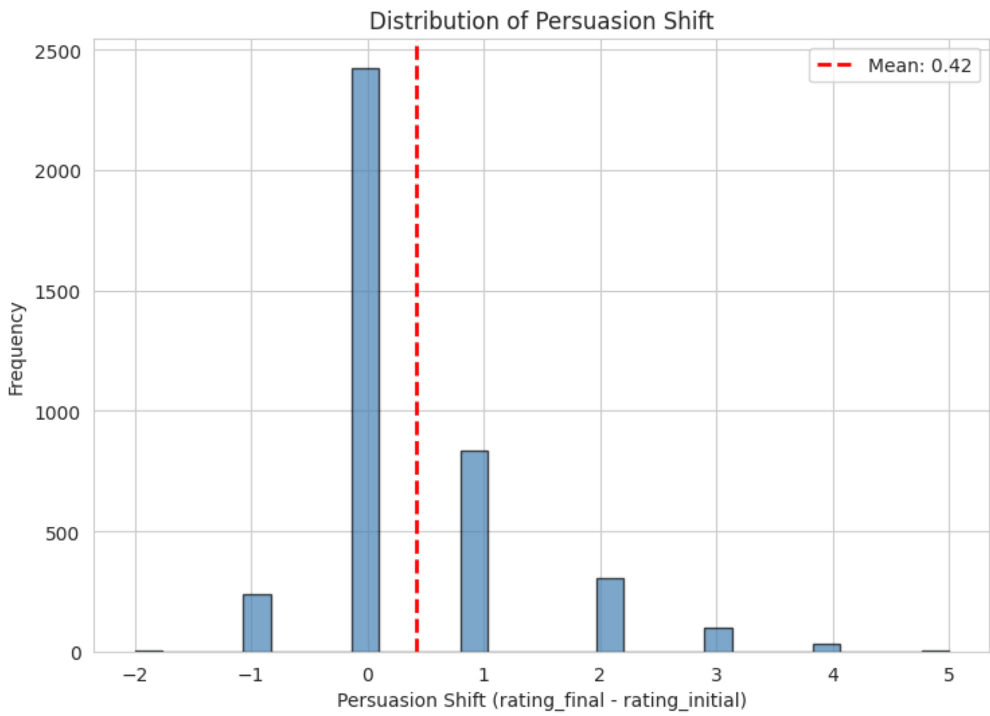


Figure 1: Distribution of Persuasion Shift

Figure 1 shows that persuasion effects tend to be clustered at zero, thus most participants maintained their rating after reading the argument. The average effect is positive at about 0.42, indicating that, on average, there will be some movement toward agreement of the scores, but the movement is not large, given the social distance shown. From a report argument perspective, this is relevant because it demonstrates that AI and Human argument may impact attitudes, however not only a slight change occurs. However, the small number of larger positive and negative shifts indicates that persuasion may not be uniform for both groups of participants and claims, which validates the second group and modelling analysis.

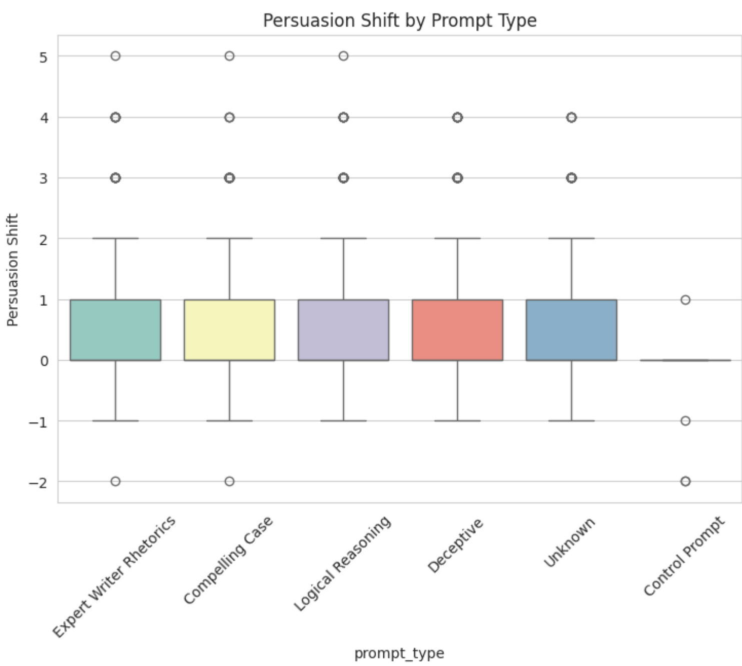


Figure 2: Persuasion Shift by Prompt Type

Figure 2 indicates that the distributions of the keys in each of the prompt-type are generally similar. Medians for Expert Writer Rhetorics, Compelling Case, Logical Reasoning, Deceptive, Unknown and Control Prompt are close to zero and most interquartile ranges are between no change and a small positive change. The scatter plots of this book's prompt types indicate that in some cases, a particular prompt was very persuasive, with high positive outliers for largely of the various prompt types, but there is no evidence of any single prompt being consistently more effective. This means that a cautious interpretation is that the promotion of persuasion involved in prompting style is compatible but it will not prevail over participant starting attitudes or claim context.

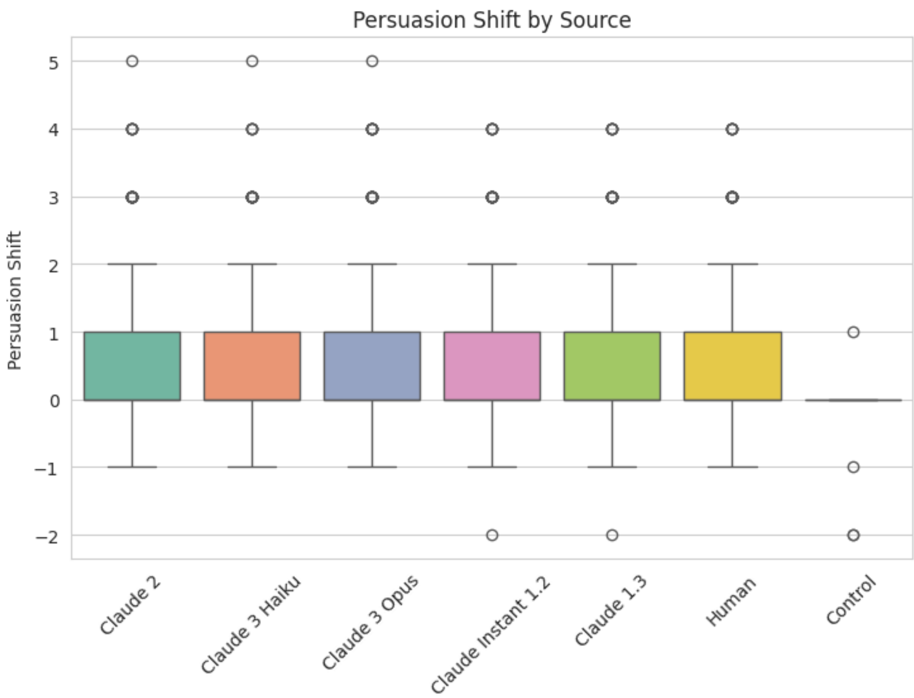

Figure 3: Persuasion Shift by Source

Figure 3 shows considerable overlap between the distributions for Claude 2, Claude 3 Haiku, Claude 3 Opus, Claude Instant 1.2, Claude 1.3, and Human arguments. Comparable central spreads and most medians in the vicinity of zero indicate that the short-term efficacy of the change in persuasion resulting from the arguments appears to be similar across sample for both the human and AI written arguments. The Control condition is more narrowly distributed about the zero score as might be expected, as there is not much of a persuasive treatment. The figure holds up a counter argument that AI can match human persuasiveness, but indicates that there are sources where it is more effective and others where it is less.

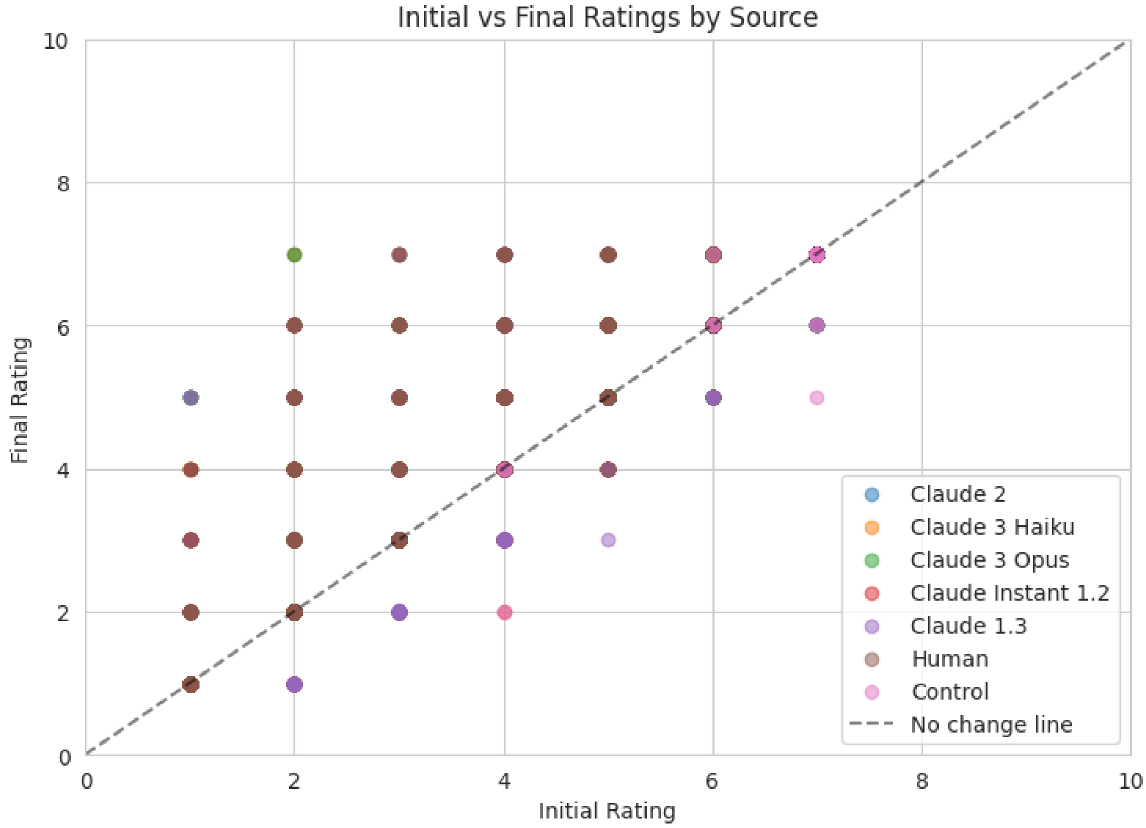


Figure 4: Initial vs final ratings by source

In Figure 4, the dashed diagonal represents no change in the initial and final rating. Points above this line reflect the rise in agreement since viewing the argument, and the points below this line reflect the decrease in agreement. Numerous observations fall on or near the diagonal, providing some credence to the limited overall persuasion effect as described in Figure 1. But some values lie above the line, particularly in the lower and middle end of initial ratings suggesting a margin for movement by those who started off having fewer positive ratings. This pattern justifies the choice of initial stance as an important covariate in the modelling phase and the importance of understanding the meaning of persuasion as being contingent on actual audience disposition instead of just source or prompt design.


Figure 5: Correlation matrix showing relationships between extracted text features and persuasion metrics.

Figure 5 shows a strong correlation (0.89) between initial and final ratings, indicating that participants' agreeability is consistent throughout the process. Those more agreeable initially remained so post-argument, while less agreeable individuals maintained their stance. This suggests that persuasion is limited by audience predisposition. Additionally, both final ratings and persuasion shift show moderate positive correlation (~0.30), while initial ratings correlate negatively (~-0.17) with persuasion shift, indicating that those with lower initial ratings had more potential for attitude change. In contrast, various text-based features showed negligible correlation with persuasion shift, implying that individual characteristics are more significant. Although word count is highly correlated with argument length (0.90), this relationship does not indicate argument effectiveness.

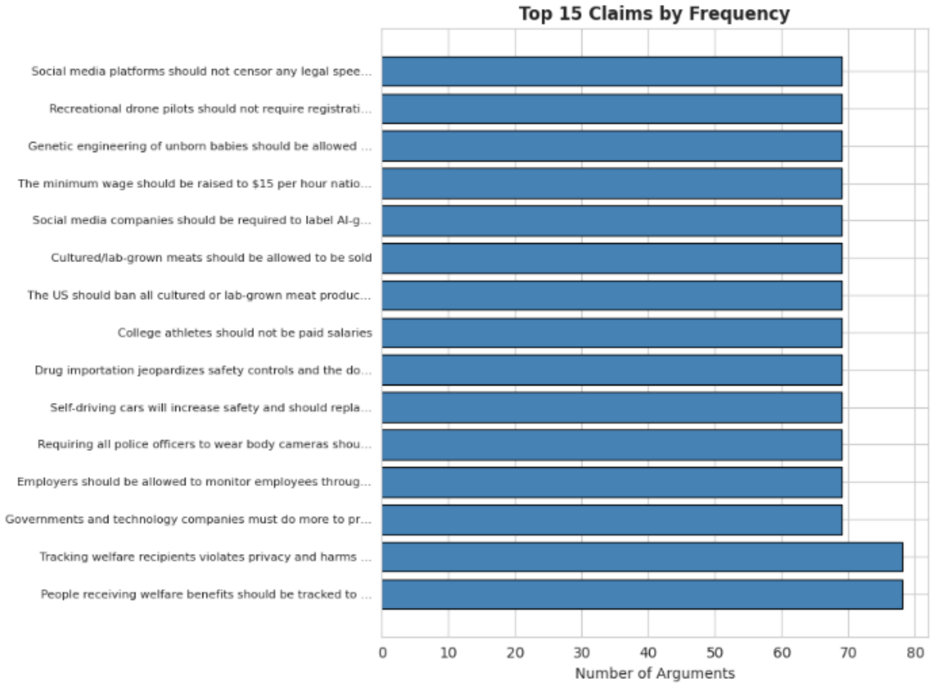


Figure 6: Top 15 claims based on the Number of Arguments

Figure 6 reveals that the most frequent claims contain a similar number of arguments (68-78), indicating a balanced distribution of data across these claims. Key areas of high frequency include welfare, social media censorship, recreational drones, genetic engineering, minimum wage, AI-generated content labels, cultured meat, college athlete pay, drug importation, self-driving cars, police body cameras, and employee monitoring. This uniformity suggests that although claim frequency may influence comparative outcomes in persuasion, the even representation allows for fair comparisons across various topics.

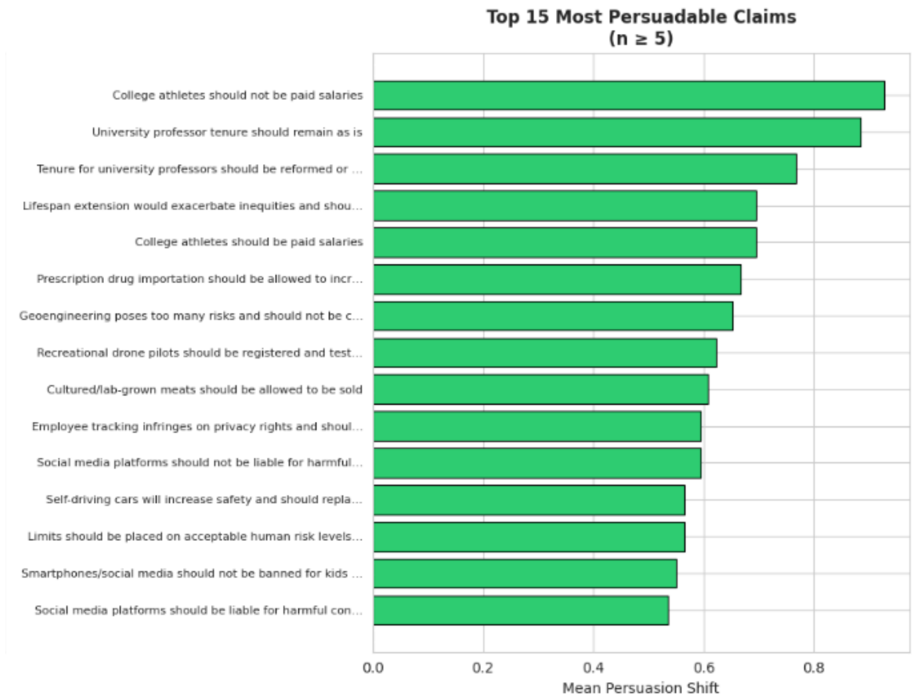


Figure 7: Top 15 Most Persuadable Claims based on Persuasion Shift

Figure 7 indicates that the most persuadable topics include college athletes' pay, university professor tenure, and other social issues. The claim "College athletes should not be paid salaries" shows a mean persuasion shift of approximately 0.9, reflecting a moderate increase in persuasion among participants exposed to both sides. This supports the project's conclusion that persuasion varies by topic, with some claims being more susceptible to change due to their less ideological nature and policy-specific focus.

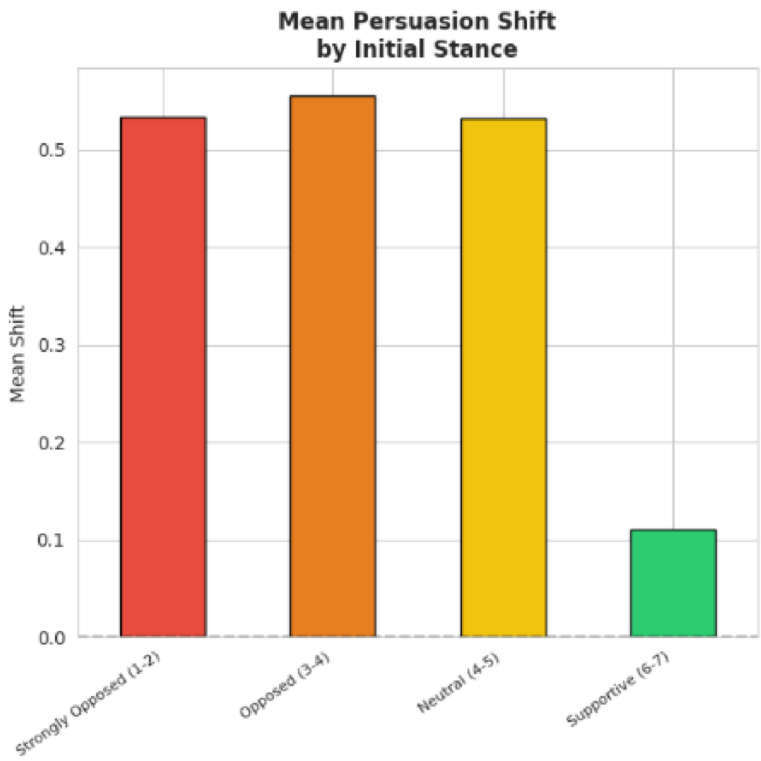


Figure 8: Mean Persuasion Shift by Initial Stance

Figure 8 shows that mean persuasion changes were largest for participants who strongly opposed or opposed, around 0.5 or slightly more. The mean shift is much less, about 0.1, for the supportive group. From this pattern, it can be determined that one of the most effective persuasion strategies was on participants who were not already in favor of the claim. Opposed and neutral parties had larger areas to reconcile their views; supported parties were nearer to the extreme end of the scale. The reason for the difficulty in predicting persuasion shift using text features only is explained in Figure 8 in particular. It may well be that the same argument has different consequences for the participant, depending on his/her initial position.

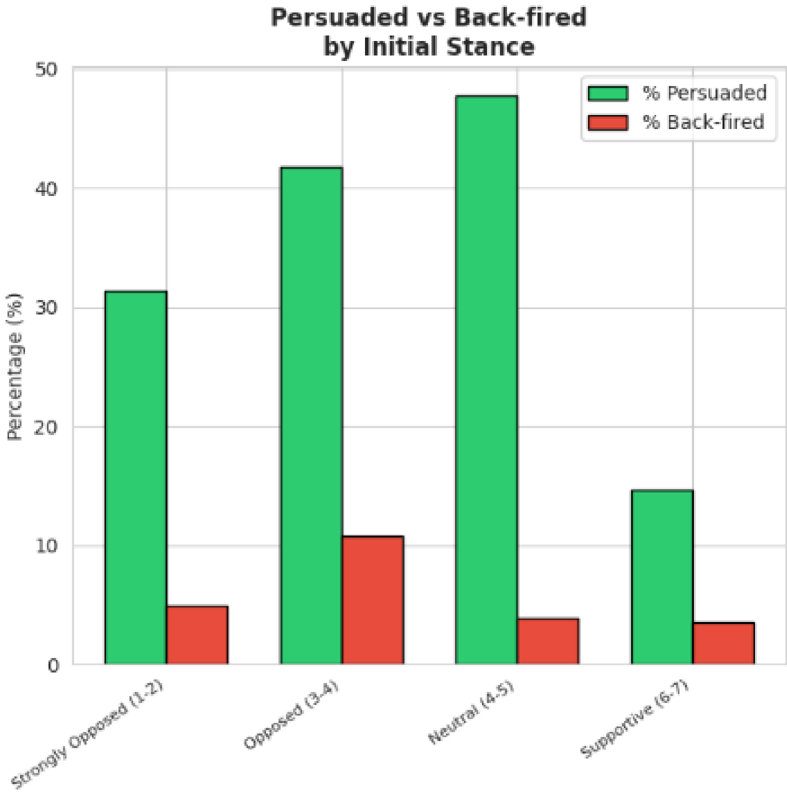


Figure 9: Persuaded vs Back-fired by Initial Stance

Figure 9 shows that the neutral group had the largest percentage of participants persuaded, followed by the opposed group, and the strongly opposed group had the lowest. About 15% in the supportive group were convinced. This is consistent with what is shown in Figure 8: those who were positive or positive but neutral were more likely to become positive; and those who were neutral or negative were more likely to become positive. All the percentages for backfire are lower than those for persuasion for all stance groups. The other groups outperform the lowest one, at about 11%, in terms of backfire rate. The size of the speech effect was important, since it indicates that persuasion was only light, but that arguments tended to more strongly sway the participants in the direction intended than to cause the opposite result.

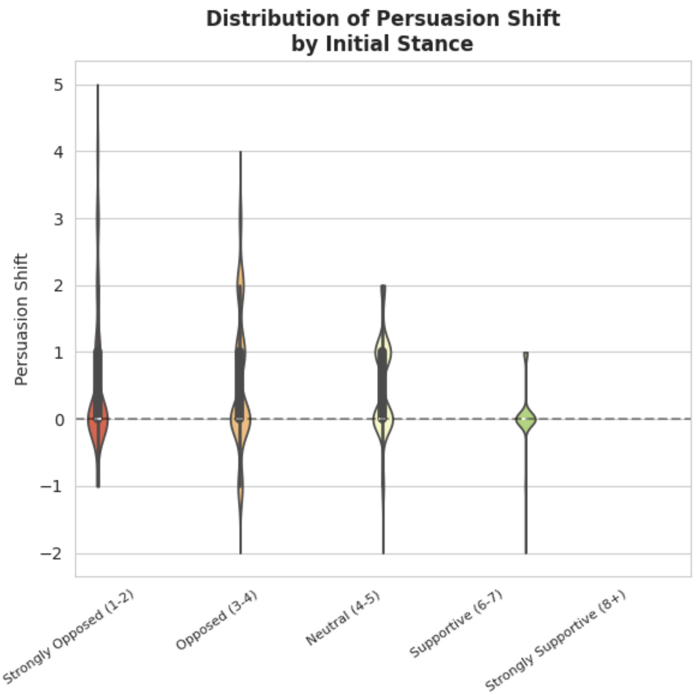


Figure 10: Initial Rating Segment Analysis

Figure 10 indicates that most persuasion shifts are centered around zero, showing that many individuals did not change their stance after reading the argument. However, the distribution type varies by group: strongly opposed and opposed groups exhibit broader peaks at the upward end, indicating significant shifts for some within these categories. The neutral group shows a positive shift with a narrower range, while the supportive group displays minimal movement and a strong focus on zero. This data supports the report's claim that understanding the audience's perspective is crucial for effective persuasion, as supportive participants had less capacity for change compared to opposed or neutral individuals.

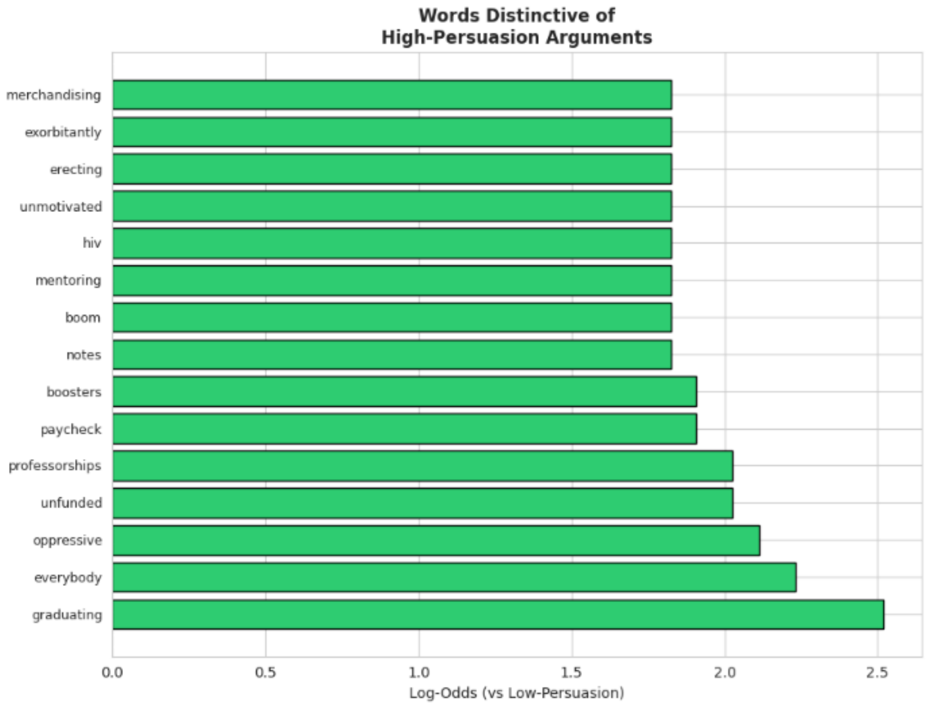


Figure 11: Words Distinctive of High-Persuasion Arguments

Figure 11 indicates that words like “graduating,” “everybody,” “oppressive,” “unfunded,” “professorships,” “paycheck,” and “boosters” correlate with high-persuasion arguments, highlighting themes of fairness, social impact, education, and employment. Successful persuasion often stems from emotionally appealing social issues or concrete examples rather than abstract reasoning. However, the association of these words with persuasion should be viewed cautiously, as they may reflect more persuadable topics rather than a direct cause of persuasion. For example, there is a connection between "graduating," "professorships," and education-related claims, while "paycheck" relates to labor or wage discussions.

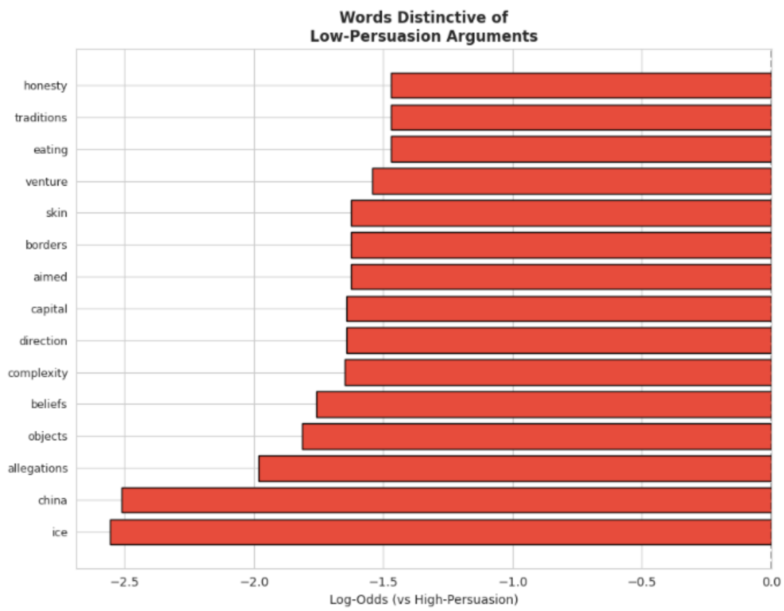


Figure 12:Words Distinctive of Low-Persuasion Arguments

Figure 12 identifies low-persuasion words such as "ice," "china," "allegations," "objects," and others, suggesting their association with complex geopolitical, immigration, and cultural issues. These terms may be less effective in facilitating short-term attitude changes due to strong pre-existing beliefs and intricate moral or political arguments. It argues that the effectiveness of persuasion relies on the interplay between vocabulary, topic, and audience predisposition, rather than the inherent value of the words themselves.

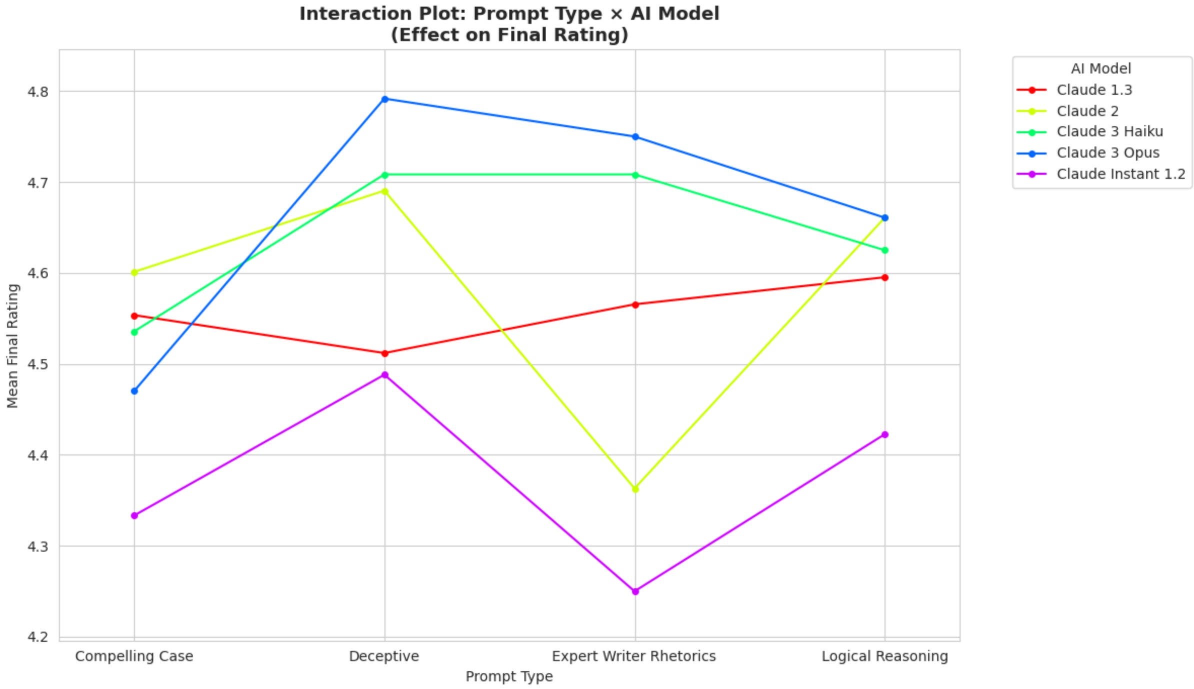


Figure 13: Interaction Plot based Prompt Type x AI Model

Figure 13 illustrates that final ratings are influenced by the AI model and prompt type employed. Claude 3 Opus achieves high mean ratings under the Deceptive and Expert Writer Rhetorics prompts, while Claude 3 Haiku performs better across various prompts. In contrast, Claude Instant 1.2 tends to have lower ratings. Claude 2 shows mixed results: strong in Expert Writer and Logical Reasoning prompts but weaker in Deceptive Rhetorics. This interaction emphasizes that prompt effectiveness varies per model, indicating that AI persuasiveness is not a universal skill but rather depends on the interplay between prompting strategies and model capabilities.

## CHAPTER 4: ETHICAL ISSUES

The Anthropic Persuasion dataset is categorized as ethically low-risk since it is a secondary dataset, devoid of human-subject data and direct identifiers such as names or email addresses. Despite being de-identified, the potential for re-identification necessitates careful handling, especially in regard to sensitive and aggregated participant identities. Hertfordshire University clarifies that studies using secondary data without participant recruitment or social media scraping do not require ethics approval, though secondary research may still require supervisor confirmation. The dataset's legal standing is affirmed, with usage under CC BY-NC-SA 4.0 for non-commercial purposes and academic analysis deemed free. Ethical considerations arise from the absence of consent forms and ethics review statements, raising questions about the dataset's trustworthiness and the validity of anthropogenic claims derived from potentially fabricated information. Consequently, any analysis involving this dataset must critically assess its ethical provenance and acknowledge the uncertainties involved.

# CHAPTER 5: METHODOLOGY

## *5.1 Research Design and Technical Workflow*

My research employed a quantitative, explanatory approach to assess whether AI-generated arguments could match or exceed human persuasive abilities. Persuasion was defined as a shift in attitude measured by participants' stance scores pre- and post-argument. The study utilized the Anthropic Persuasion dataset, which included claims, arguments, sources, prompt types, and persuasiveness ratings, enabling both regression and classification analyses. The technical workflow comprised four modules: data loading, cleansing, rating conversion, and feature engineering, followed by exploratory analysis, statistical testing, model training, and evaluation, all aimed at ensuring consistency in the research question and methodologies presented in the results chapter.

## *5.2 Data Preparation and Feature Engineering*

To preprocess the data, categorical stance ratings were transformed into a numeric scale from 1 to 7, where higher scores indicate support. The change in ratings was calculated to measure shifts in persuasion, categorizing outcomes as increases, neutralizations, or decreases in agreement. Records with improperly mapped ratings were deleted, and missing ratings were labeled as "Unknown." Text was normalized by lowercasing, removing punctuation, and standardizing whitespace. Contextual features such as Source, Source type, Prompt type, and Initial rating, along with textual features like emotion-word count, evidence-marker count, argument length, and sentence statistics, were considered, as the linguistic structure and reader attitude could influence persuasion effectiveness.

## *5.3 Statistical and Machine Learning Models*

Four modeling approaches were employed to establish a consistent methodology for predicting final ratings and analyzing text features. First, Linear Regression provided interpretable characteristics of features influencing persuasiveness. Next, ordinary least squares regression using the statsmodels framework assessed statistical contributions through p-values. A Random Forest Regressor with 100 trees was then utilized to capture non-linear feature interdependencies and measure feature importance. Lastly, two lightweight models optimized with QLoRA, unsloth/tinyllama-chat-bnb-4bit and unsloth/llama-3.2-1b-bnb-4bit, were used to classify a test set into persuasion shift categories: Negative, Neutral, and Positive, facilitating a comparison between traditional regression and transformer-based models in the persuasion task.

## *5.4 Hypothesis Testing and Metrics*

I investigated differences in final ratings and persuasion changes based on source, prompt type, textual features, and initial stance using statistical methods. Descriptive statistics, correlation analysis, t-tests, and ANOVA models facilitated group comparisons, while regression evaluation incorporated Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and R-squared metrics. These metrics were suitable given the continuous numeric nature of the final ratings. The analysis revealed that the root mean square error penalizes larger errors, and R-squared indicates the variance proportion accounted for by the model. For purpose shift classification, I used accuracy and macro F1 score, noting that macro F1 is more appropriate for comparing the Negative, Neutral, and Positive classes due to its equal weighting.

### *5.5 Extension Modelling*

I extended the method by researching persuasion in the initial-stance segments: strongly opposed, opposed, neutral, and supportive participants, as well as strongly supportive participants. This was significant since not all readers may be convinced the same way. There was less headroom for those that were already with a claim, while those that were opposed to it or neutral had more headspace for a positive persuasion shift. I also added a comparison of prompt strategies and AI-sources to look at the performance and determine if there was a preference between certain types of prompts and sources for specific models. The fact that this extension identified when persuasion occurred, which variables were more predictive of persuasion, and which segments were most likely to be persuaded helps connect the predictive modelling to the main research question of when it would and could not occur, as well as which segments were most susceptible to persuasion.

# CHAPTER 6: RESULTS

## *6.1 Choosing the Most Appropriate Metrics*

Regression models were better in predicting final persuasion ratings than changes. An OLS/Linear Regression model with MAE = 0.6654, RMSE = 0.8727, and $R^2$ = 0.7901 accounted for roughly 79% of the variance in final scores. The triangles follow the line denoting a perfect fit in the first figure, but the prediction models seem to smooth each actual rating, thus what is visible is smoothing error. With an average $R^2$ of 0.945, the Random Forest regressor outperformed the OLS in modelling non-linear relationships. Its figure shows projected values near to pattern values, especially at higher rating levels (Karinshak et al., 2023). While attitudes toward persuasion shift were easier to predict, the R2 value of 0.033 was low, limiting its ability to explain attitude shifts. TinyLlama's classification accuracy matches LLaMA-3.2's 0.6389 and macro F1's 0.2599. Although correct, balanced class performance was poor, especially in Negative, Neutral, and Positive shift categories. LLaMA-3.2 was more efficient since it reduced training time while maintaining performance.

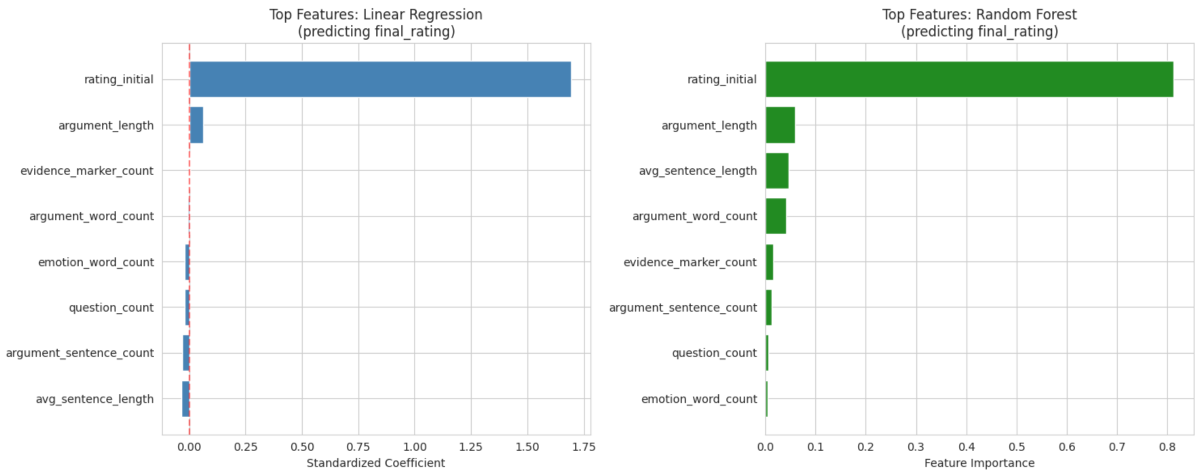

Figure 14: Top predictive features for final rating from linear regression (standardized coefficients) and random forest (feature importance)

## *6.2 Best Ways to Present the Results*

Multiple visualisations of persuasion distribution and causation supported the numerical findings. A histogram of the persuasion shift indicates a strong central tendency with a mean shift of +0.415 and a standard deviation of 0.887, showing that 82.66% of arguments did not affect belief. Boxplots were used to compare the mean shifts for persuasion by human and AI arguments and the mean shifts for specific prompt types within AI arguments. Human arguments had a mean shift of 0.480, AI arguments 0.404, and "Deceptive" 0.470, which was higher than "Infectum" (0.239) and "Warn" (0.454) (Barale, 2021). To demonstrate this strong linear link between the two variables, a scatter plot was generated with most data points near the line "no change" for the initial and final ratings. Additionally, Random Forest model-based feature importance bar charts showed that the first rating is the most essential feature, encompassing 0.812 of feature importance, far more than other textual features like argument length (0.059). To support the visual records with statistical data, correlation heatmaps showed a substantial positive correlation between initial and final scores (0.888) (Ruiz-Arellano *et al.,* 2022).

## *6.3 What the Results Mean in This Project*

The empirical results substantially change computer-based persuasion ideas. According to descriptive data, the average argument only shifts 0.415 on a seven-point belief scale, indicating that most arguments don't modify established beliefs. The OLS regression analysis reveals a significant coefficient of initial rating (1.694; $p < 0.001$), indicating that textual features like argument length and evidence markers had little impact on this weakness (Ruiz-Arellano et al., 2022). The argument a person makes will be influenced by their previous thoughts, even if it is ineffective. The low $R^2$ of the persuasion shift model (0.033) suggests that additional factors, not just text qualities, contribute to the variance between pre- and post-states. The ANOVA results indicated that prompt types had a substantial impact on final evaluations ($F = 15.31$, $p < 0.001$). Transformer models, with about 0.63 accuracy, show that machine learning can now pick up general persuasion patterns, making the nuanced, three-way shift in human opinions partially solved (Karinshak *et al.,* 2023).

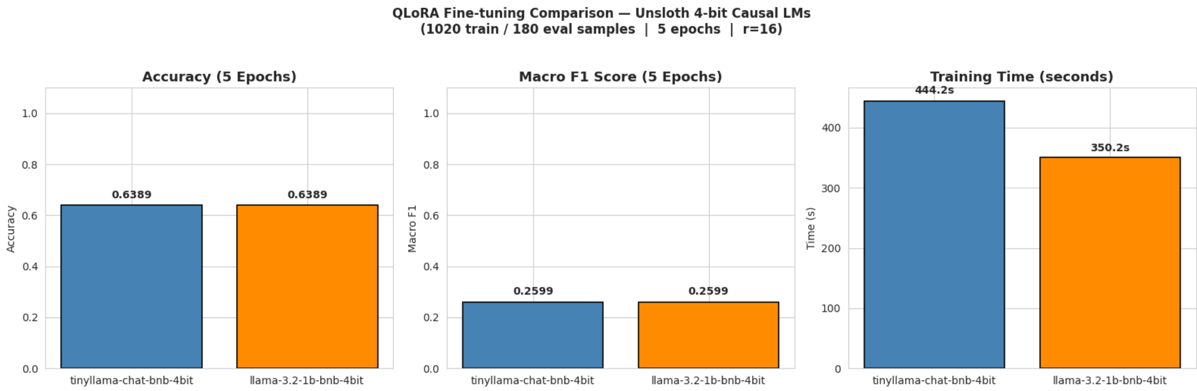


Figure 15: Model Comparison between TinyLlama and LLaMA-3.2 fine-tuned models

### *6.4 Implications for Real-World Use*

Real-world ramifications arise from quantitative AI-human comparisons. The mean final rating for human written arguments (4.877) significantly surpassed AI written arguments (4.564) with P value < 0.0003, according to an independent t-test. This lower figure suggests that while big language models have improved, AI has yet to fully mimic human persuasive language. The dataset also indicated that the "Deceptive" question (mean shift = 0.470) and "Logical Reasoning" prompt (mean shift = 0.445) had mean shifts that were close to or better than other AI baselines. Fine-tuning a transformer like TinyLlama and DistilBERT to classify persuasion shifts in a text is more than 60% accurate, suggesting automated auditing or text generation (Huang and Wang, 2023). These results cast doubt on AI's general dominance over human persuasion, but with training, coaching, and targeted prompting, AI can methodically frame discourse. To prevent automation-driven, huge manipulation, new and robust policies, digital media principles, and public health measures and capacities are needed.

### *6.5 Addressing the Research Question*

| Model | Task Type | Target Variable | MAE | MSE | RMSE | $R^2$ Score | Overall Interpretation |
|---|---|---|---|---|---|---|---|
| Linear Regression | Regression | Final rating | 0.6654 | 0.7615 | 0.8727 | 0.7901 | The Linear Regression model performed well and explained 79.01% of the variance in final rating. It was useful for |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | interpretation because it showed the relationship between initial rating, text features, and final persuasion rating. |
| Random Forest Regressor | Regression | Final rating | 0.2976 | 0.1999 | 0.4471 | 0.9449 | The Random Forest model achieved the strongest regression performance, with lower error values and a higher $R^2$ score than Linear Regression. This suggests that non-linear feature interactions improved prediction accuracy. |

| **Model** | **Task Type** | **Target Variable** | **Accuracy** | **Macro F1** | **Training Time** | **Training Loss** | **Overall Interpretation** |
|---|---|---|---|---|---|---|---|
| unsloth/ tinyllama-chat-bnb-4bit | Classification | Persuasion shift class: Negative, Neutral, Positive | 0.6389 | 0.2599 | 483.8 seconds | 0.8297 | TinyLlama achieved 63.89% accuracy, but the low macro F1 score suggests weaker balanced performance across the three persuasion-shift classes. |
| unsloth/ llama-3.2-1b-bnb-4bit | Classification | Persuasion shift class: Negative, Neutral, Positive | 0.6389 | 0.2599 | 333.9 seconds | 1.3871 | Llama 3.2 1B achieved the same accuracy and macro F1 as TinyLlama but trained faster. However, its higher training loss suggests less efficient learning compared with TinyLlama. |

The study elucidates that initial stance, final rating patterns, claim topic, and feature interactions significantly influence persuasiveness more than text length. Results from various figures indicate weak correlations between argument length, emotion-word count, and persuasion shift, while initial and final ratings show a strong correlation of 0.89. The study confirms that audience openness is crucial, as indicated by variations in persuasion shifts between initially opposed and supportive members. Additionally, while distinctive words indicate differences between high and low persuasion arguments, rhetorical features alone do not account for persuasion variations. Claude 3 Opus prompt type yields higher final ratings, supported by a Random Forest model with $R^2 = 0.9449$, indicating strong regression performance.

# CHAPTER 7: ANALYSIS AND DISCUSSION

### *7.1 Meaning of the Results*

The analysis indicates that while there is significant persuasion in the project, it is limited, with findings showing only minor shifts in responses post-exposure to arguments. This reflects a more genuine trend in potential attitude change, influenced heavily by the reader's initial rating, which acts as the strongest predictor of final ratings (Huang and Wang, 2023). The analysis indicates that while there is significant persuasion in the project, it is limited, with findings showing only minor shifts in responses post-exposure to arguments. This reflects a more genuine trend in potential attitude change, influenced heavily by the reader's initial rating, which acts as the strongest predictor of final ratings.

### *7.2 Which Model Works Best and Why*

Linear regression is more interpretable while Random Forest excels in predictive performance, particularly in persuasive contexts. Linear regression estimates linear relationships through the minimization of squared residuals, revealing the contributions of predictors like source and prompt (Soliman, 2024). Linear regression is more interpretable while Random Forest excels in predictive performance, particularly in persuasive contexts. Linear regression estimates linear relationships through the minimization of squared residuals, revealing the contributions of predictors like source and prompt (Dehnert and Mongeau, 2022).

### *7.3 Comparison with the Literature*

The project's findings are consistent with the literature discussed above, but more conservative than recent media assertions about AI-persuasion studies. According to Anthropic, model generation boosted persuasiveness and the Claude 3 Opus did not significantly differ from human-written arguments in their benchmark. Lin et al. found pre-registered trials suggesting that AI chats could affect voter opinions more than standard video ads (Karinshak et al., 2023). These works suggest AI can be objectively convincing. The current project appears less brazen, and persuasion relies on previous position and circumstances rather than just the source. That match is that AI-based persuasion must be seen as a communication process driven by cues, context, and human perception, not just a message delivery. It is also consistent with Millet et al. (2023), who discovered that AI-generated outputs were evaluated poorer, demonstrating that source identification can affect evaluations independent of content quality. This underground project does not contradict literature.

### *7.4 Limitations and Practical Use*

Its findings have notable flaws; first, the dataset is limited to short-term opinion changes and does not focus on long-term behavioural changes. It only represents a small part of the field of persuasion; early positions are more predictive than anticipation, which does not consider various other significant dimensions of persuasion, including rhetorical formats and emotional casing. Its generalizability is limited because the data are confined to an experimental system and do not cover all real-world cases of persuasion (Ruiz-Arellano et al., 2022). Whereas the models can foresee persuasive effects and strategies in certain circumstances, such as education or health communication, there are ethical issues associated with their application, especially in politics, where success in persuasion does not necessarily imply truthfulness.

## CHAPTER 8: CONCLUSION

In conclusion, the analysis of 3,939 records indicates that AI-generated arguments are empirically persuasive but slightly less effective than human arguments, with a mean shift of 0.415. The initial stance significantly predicts final attitude, evidenced by a high correlation coefficient of 0.888 and explaining 79% of variance in the OLS regression model. AI arguments produce a mean shift of 0.470 compared to a human baseline of 0.877, with a significant p value of 0.0003. Specific AI prompts like “Deceptive” reasoning show competitive results. Persuasion is complex, hinging on the recipient's views rather than just the facts presented. These findings have practical implications for health communication and policy messages. Future research should explore advanced linguistic features and the impact of immediate rating changes on long-term behavior outcomes.

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# APPENDIX